

Chapter 11

The Determination of Aggregate Output, the Price Level, and the Interest Rate

Lecturer: Mihyun You

Department of Economics
Georgetown University

Summer 2024



GEORGETOWN UNIVERSITY

Contents

- 1 The Aggregate Supply (AS) Curve
- 2 The Aggregate Demand (AD) Curve
- 3 The Final Equilibrium



GEORGETOWN UNIVERSITY

Recap: The Macroeconomic Theory So Far

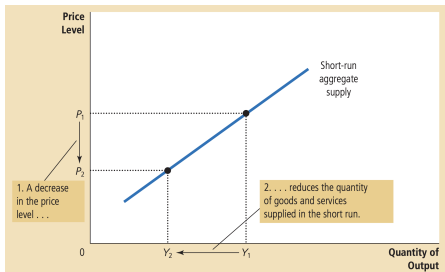
- In Chapters 8 and 9, we looked at how the output level in the economy is determined, keeping the interest rate and the price level fixed.
- In Chapter 10, we turned our attention to the interest rate, holding output and the price level fixed.
- We are now ready to bring together these pieces of the economy—output, the price level, and the interest rate.

Key to Understanding AS Curve

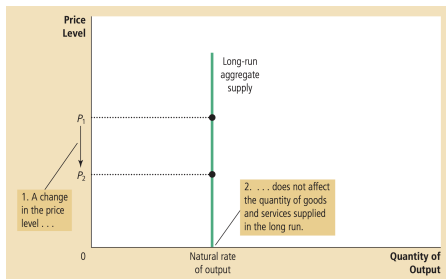
- The aggregate supply (AS) curve shows the relationship between the aggregate quantity of output supplied by all the firms in an economy and the overall price level.
 - Although it is called an *supply* curve, it is somewhat misleading.
 - It is better thought of as a "price/output response" curve.
- Economists distinguish between long-run and short-run according to whether the economy is in full employment or not.
 - Long-run: prices and wages are flexible, labor and capital are fully adjustable. Expected inflation is same as the actual inflation.
 - Short-run: prices and wages are sticky, capital is hard to adjust.

The Short-Run and Long-Run AS curve

- AS curve represents different shapes depending on time horizons.
 - Long-run AS curve: vertical line
 - Short-run AS curve: upward sloping



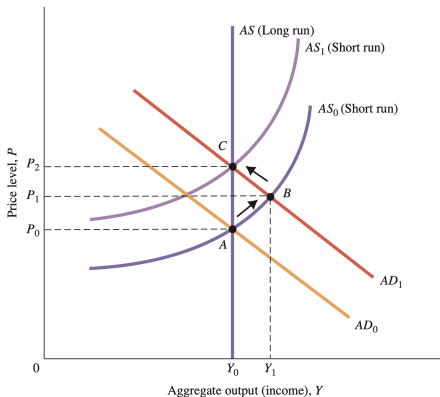
Short-run AS Curve



Long-run AS Curve

Potential GDP

- Potential output $\bar{Y}$: the level of aggregate output that can be sustained in the long run without inflation.
 - Synonyms: long-run level of output, natural level of output, full-employment output

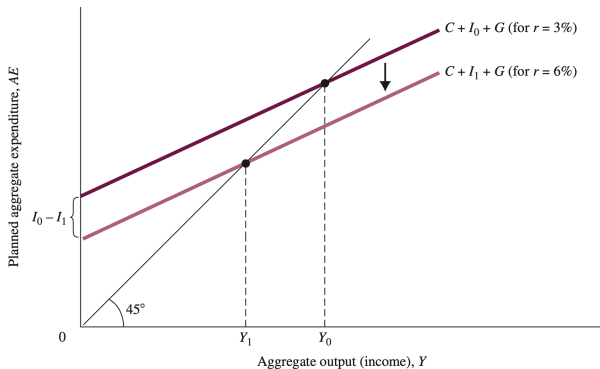


Adding interest rate (r) into the Keynesian Model

- Planned Aggregate Expenditure and the Interest Rate

$$AE = C(Y) + I(\textcolor{red}{r}) + G$$

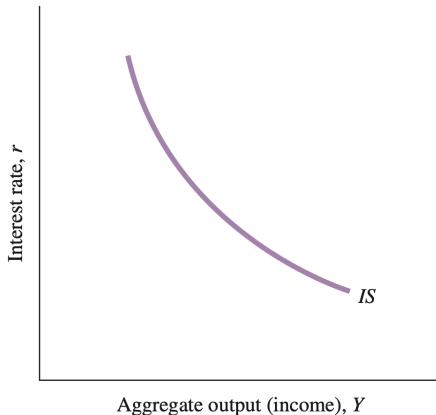
- A high interest rate (r) discourages planned investment (I).



IS Curve

- *IS* curve represents equilibrium in the goods market for the particular interest rate.
- Equilibrium means that planned investment equals saving, hence the *IS* notation.
- To trace out an *IS* curve, start with a Keynesian cross with *a given interest rate*, then *change the interest rate*.

IS Curve



! What does *IS* curve look like when I does not depend on r ?

LM Curve

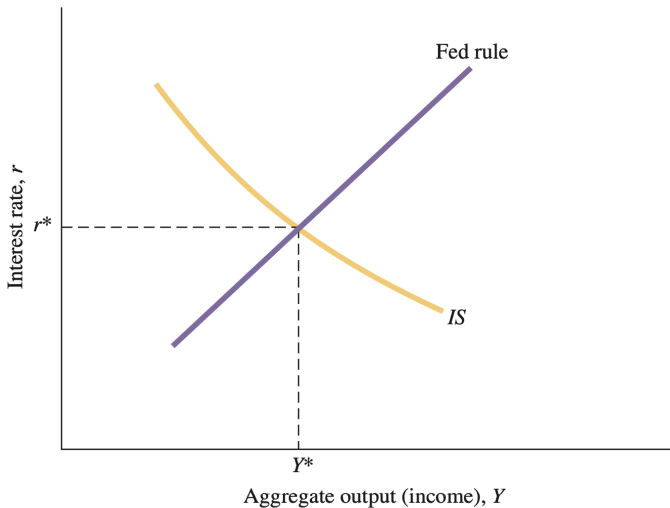
The Behavior of the Fed

- How is the equilibrium interest rate determined?
- The Fed controls the interest rate.
 - Every six weeks, the Federal Open Market Committee (FOMC) has a meeting to decide on the value of the federal funds rate.
- Fed rule: equation that shows how the Fed's interest rate decision depends on the state of the economy

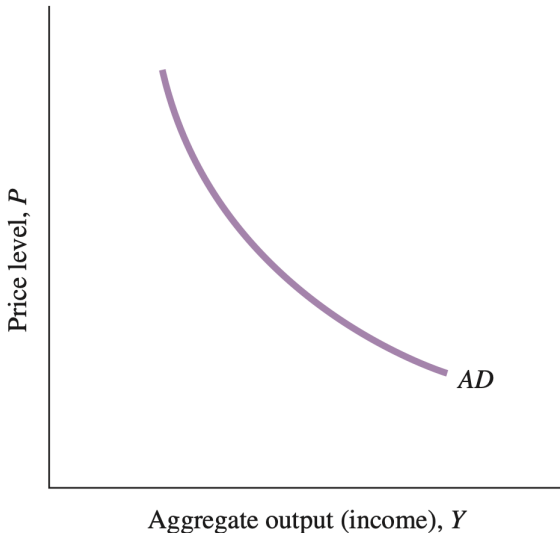
$$r = \alpha Y + \beta P + \gamma Z$$

where $\alpha, \beta, \gamma > 0$.

$IS - LM$ Model



Deriving the *AD* Curve



Equilibrium Output and the Price Level

